Smart Student Attendance System Using QR Code

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Abstract- **QR Code has a wide range of applications in this evolving technology world. QR code used to store massive information in a smaller space. So we decide to use QR code in our system and proposed smart attendance system using QR code. Secure authentication, is achieved using data-hiding algorithms with the embedded QR Code. In our project by using smartphones student scan QR code which will displayed by the teacher. When student scan this QR code, automatically attendance will be marked according to the user id. It also discusses how the system verifies student identity to eliminate false registrations.**

Keywords- **Mobile Computing; Database; Attendance System.**

I. INTRODUCTION

Taking attendance using paper and pen was one approach we could have used, but we knew it was slow and prone to errors. In addition, the paper method required a data entry phase in order to generate reports, which also suffered from similar problems. Now days most of the people have android smart phone so we developed one android application and with the help of this application Student, scan use web application generated QR code. It is very useful for a, Students to check their attendance. So that, the student will come to know how much he/she had got the attendance presentence in one month and now how much attendance he should get to present next month. In this our project Student login into the android application. QR Code is generated in server side by teacher and student scan QR code use in android application. Student get notify and view attendance info in android application. In this system Teacher login into the web application. Calculate monthly attendance by System and also report to student parents about less attendance with mail process if student has less than 50% attendance.

II. PROPOSED APPROACH

The proposed system provides that instead of taking attendance of a sheet of a paper teacher can use QR code to take attendance of the student. The system is divided into two part teacher module and student module. Where teacher module is a web application and student module is an android application. In a teacher, the module teacher can generate QR code, view and edit attendance as per student, calculate monthly attendance and send notification for the student. In a student, module student can scan QR code to mark attendance for that lecture. A student can also view his/her attendance and view notification send by the teacher. We are going to used API to generate QR code in our project. All the data is stored in MYSQL database so that database backup can be easily done. By taking all this point into consideration we had tried to overcome the limitation of the previous system and try to make our proposed system which will save the time of teacher and student and which will avoid usage of paper by which we can save a lot of paper and teacher doesn't have to maintain any record as the data get stored directly in the database.

A] A QR Code is a two-dimensional barcode that is readable by smartphones. It allows to encode over 4000 characters in a two dimensional barcode. QR Codes may be used to display text to the user, to open a URL, save a contact to the address book or to compose text messages. The system requires a simple login process by the class instructor through its Server Module to generate an QR code with specific information. During the class, or at its beginning, the instructor enter the lecture information and how much time required to display the QR image. The QR image is then generated and the students can then scan the displayed QR code using the system Mobile Module through the smartphone, the Mobile Module will then communicate the information collected to the Server Module to confirm attendance. The whole process should take less than a minute for any student as well as for the whole class to complete their attendance confirmation. Smartphones may communicate with the server via either the local Wi-Fi coverage offered by the institution or through the internet.

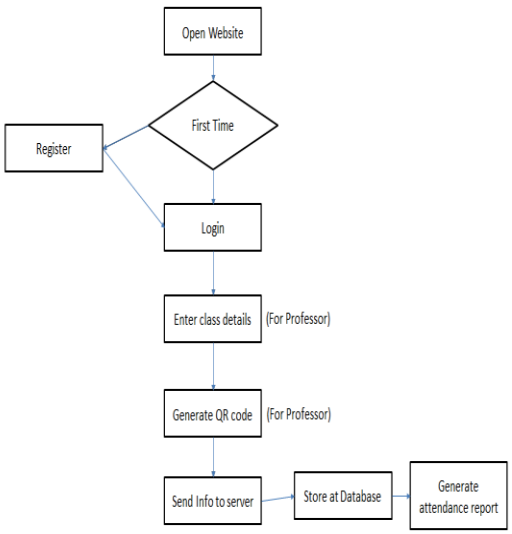


Fig.1

In figure 1 show the Flow chart shows the process on how the scenario will be played out.

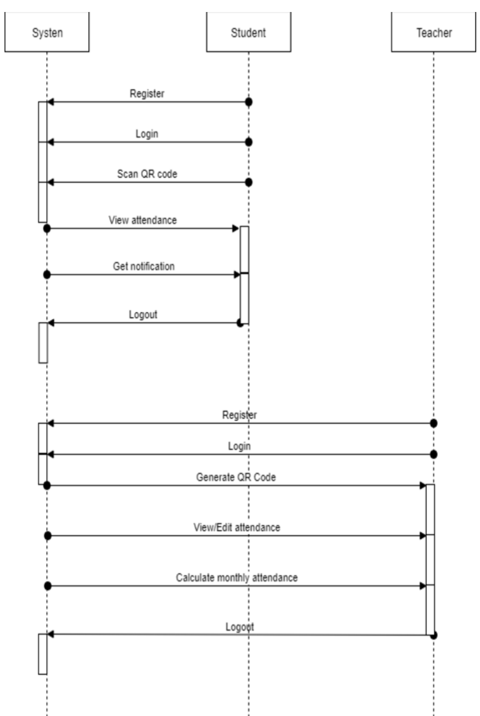


Fig 2

For clear understanding , use case in fig 2 can help you to understand the clear working of it.

B] The QR code algorithm is based on number of parameter that will make the QR code unique. This procedure will aid the faculty in reducing the chances of same QR code appearing. To make the QR code unique we pass-lecture -name +current date +system-current-millisecond +UUI method that is 15 variables long which random and unique.

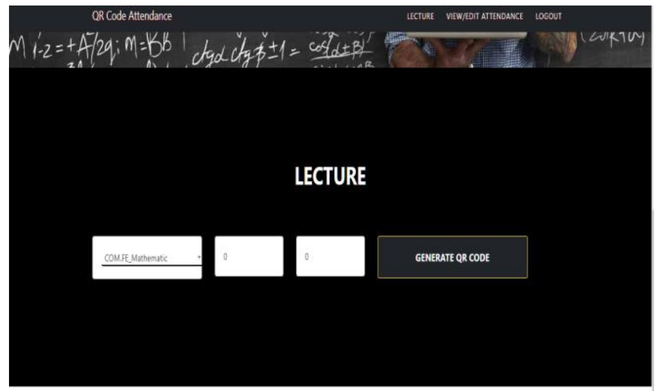


Fig.3

In figure 3 show the website part for the teacher to generate QR code.

Fig.4

Figure 4 show the QR code generated for the students.

QR code Algorithm is as follows:

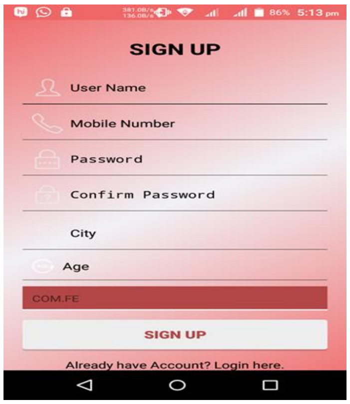
- Create a variable to store the text for which we want to generated qr code
- Create a ByteArrayOutputStream and pass QrCode.from() method which take variable in which the text is stored. along with this you can used to() and withSize() method where to() method take image type(png or jpg) and withSize() you can give size to qr code and it manditory to used stream() method along with it as ByteArrayOutputStream only to store stream so we can convert it into stream using stream() method.
- Create byte[] array and pass ByteArrayOutputStream object but first convert it into byte array using toByteArray() method so that it can be stored.
- Qr code is genterated

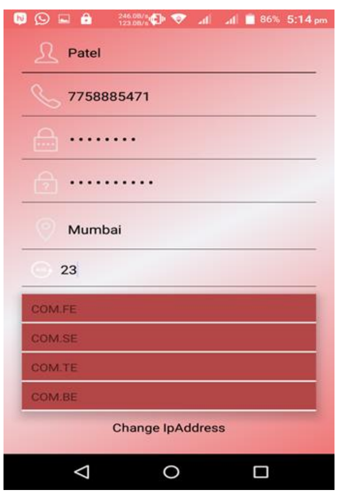
- We can save that qr code in database by passing byte array

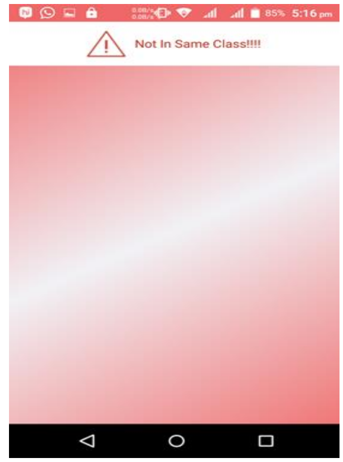
This flow we find out from three jar files which are

- zxing-core-1.7.jar
- zxing-javase-1.7.jar
- QRGen.jar

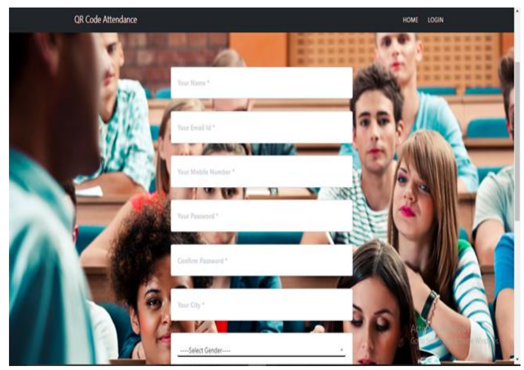
RESULTS OF THE SYSTEM ANDROID PART

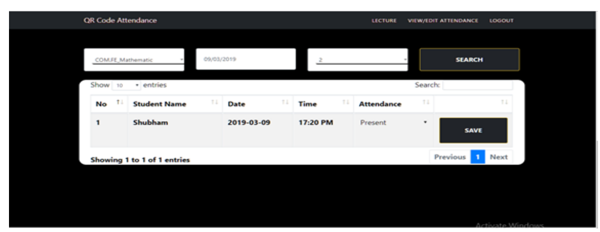






RESULTS OF THE SYSTEM WEBSITE PART





III. AVAILABILITY

The system will be available 24x7, as it will be a web based portal.

IV. MAINTAINABILITY

The system would be maintainable to a good extent since there are not too much of the hardware devices. The system is solely based on internet connectivity, and database has to be maintained against attacks and has to be dynamic in nature, where they can be modified.

V. PORTABILITY

The system is completely portable since the only requirement is internet connectivity. Also, the web page is responsive and works well with mobile, desktop PC.

VI. CONCLUSION

These days it is required to keep up with the latest technologies, especially in the field of education. Educational institutions have been looking for ways to enhance the educational process using the latest technologies. Looking at the existing situation, we have thought of using the mobile technology to efficiently benefit from the complete assigned time assigned to a lecture. Time taken by instructors to take attendance may be viewed sometimes as a waste of the lecture time, especially when classes are big. For that, we have proposed a way to automate this process using the students' devices rather than the instructor's device. In other words, the instructor need not do anything extra during the class beyond presenting the slides of the subject to be taught to the students.

VII. ACKNOWLEDGMENT

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